Reproducibility Report: Maglio & Polman (2014), Study 1

Bumper-rails version

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> **Goals**
>
> 1. Load a published dataset with `here()` and `read_excel()`
> 2. Reproduce the ANOVA and effect sizes reported in the paper
> 3. Say clearly what did and did not reproduce, and why

This is the **bumper-rails** version: most of the code is written for you, and you replace each `"FIXME"` with the right variable name or expression. If you would rather write the code yourself, use `reproducibility-report-minimal.qmd` instead. Both reproduce the same results.

`error: true` is set at the top of this file, so the document still renders while it has `FIXME`s in it. The chunks you have not finished yet show their error message in red, which is a handy to-do list.

Before you start, read Study 1 in the paper: `original_paper/maglio-polman-2014.pdf` (two pages).

The study

Maglio, S. J., & Polman, E. (2014). Spatial orientation shrinks and expands psychological distance. *Psychological Science, 25*(7), 1345–1352. https://doi.org/10.1177/0956797614530571

Methods summary. Researchers recruited 202 volunteers at the Bay Street subway station in Toronto. Half were waiting on the westbound platform and half on the eastbound platform. Each participant was randomly assigned to rate how far away one of four stations felt (1 = very close, 7 = very far): Spadina (two stops west), St. George (one stop west), Bloor-Yonge (one stop east), or Sherbourne (two stops east). The authors ran a 2 × 4 ANOVA on the distance ratings, then compared eastbound and westbound travelers separately at each station.

Target outcomes. This is the result you are trying to reproduce, quoted from the Study 1 results section:

> We carried out a 2 (orientation: toward, away from) × 4 (station: Spadina, St. George, Bloor-Yonge, Sherbourne) analysis of variance (ANOVA) on closeness ratings, which revealed no main effect of orientation, $F < 1$, and a main effect of station, $F(3, 194) = 24.10$, $p < .001$, $p^2 = .27$. This main effect was qualified by the predicted interaction between orientation and station, $F(3, 194) = 16.28$, $p < .001$, $p^2 = .20$. We decomposed this interaction by the subjective-distance ratings between participants traveling east and west for each of the four subway stations. Westbound participants rated the stations to the west of Bay Street as closer than did eastbound participants; this effect was obtained for both the station one stop to the west (St. George, $p < .001$, $p^2 = .28$) and the station two stops to the west (Spadina, $p = .001$, $p^2 = .20$). The opposite pattern held true for stations to the east of Bay Street. Eastbound participants rated the stations to the east of Bay Street as closer than did westbound participants; this effect was obtained for both the station one stop to the east (Bloor-Yonge, $p = .053$, $p^2 = .08$) and the station two stops to the east (Sherbourne, $p < .001$, $p^2 = .24$). Figure 1 summarizes these results.

p^2 is **partial eta squared**, an effect size. You will compute it below.

Step 1: Load packages

`readxl` (reads Excel files) and `broom` (turns model output into tibbles) are installed along with the tidyverse, but `library(tidyverse)` does not load them, so we load them by name.

```
install.packages(c("tidyverse", "here"))
```

```
library(tidyverse)
library(readxl)
library(broom)
library(here)
```

Step 2: Load the data

The authors posted one Excel file per study. We only need Study 1. The columns are described in `docs/codebook.md`.

```
d <- read_excel(here("data", "S1_Subway.xlsx"))

glimpse(d)
```

```
Rows: 202
Columns: 4
$ DIRECTION  <chr> "EAST", "EAST", "EAST", "EAST", "EAST", "EAST", "EAST", "EA~
$ DISTANCE   <dbl> 1, 2, 2, 3, 1, 1, 2, 1, 3, 2, 1, 1, 2, 1, 3, 1, 2, 1, 1, 1,~
$ STN_NUMBER <dbl> 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3,~
$ STN_NAME   <chr> "B-Y", "B-Y", "B-Y", "B-Y", "B-Y", "B-Y", "B-Y", "B-
Y", "B-~
```

Step 3: Tidy the data

The data are already tidy: one row per participant, one column per variable. But R has read `DIRECTION` and `STN_NAME` in as text. For an ANOVA they need to be **factors**, and we set the order of the station levels to run west to east, so tables and plots come out in map order.

```
d <- d |>
  mutate(
    DIRECTION = factor(DIRECTION),
    STN_NAME = factor(STN_NAME, levels = c("SPAD", "STG", "B-Y", "SHER"))
  )
```

Step 4: Reproduce the analysis

How many people are in each cell?

Count the participants in each direction × station combination. Replace each `"FIXME"` with a column name.

```
d |>
  count(DIRECTION, STN_NAME)
```

```
# A tibble: 8 x 3
  DIRECTION STN_NAME     n
  <fct>     <fct>    <int>
1 EAST      SPAD        26
2 EAST      STG         26
3 EAST      B-Y         23
4 EAST      SHER        26
5 WEST      SPAD        25
6 WEST      STG         25
7 WEST      B-Y         26
8 WEST      SHER        25
```

Does this match the sample size in the paper (202 total, 101 per platform)? Is the design balanced, with the same n in every cell?

The 2 × 4 ANOVA

When a paper says "analysis of variance", a good first step in R is `aov()`. The formula `y ~ A * B` asks for both main effects *and* their interaction; it is shorthand for `y ~ A + B + A:B`.

Fill in the formula with the outcome (the distance rating) and the two predictors.

```
mod <- aov(DISTANCE ~ DIRECTION * STN_NAME, data = d)

summary(mod)
```

```
                   Df Sum Sq Mean Sq F value   Pr(>F)
DIRECTION           1   0.71   0.713   0.664    0.416
STN_NAME            3  75.16  25.053  23.349 6.01e-13 ***
DIRECTION:STN_NAME  3  52.41  17.471  16.283 1.77e-09 ***
Residuals         194 208.15   1.073
```

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The numbers in parentheses in "F(3, 194)" are the degrees of freedom: 3 for the effect, 194 for the residuals. Check that yours match.

The teaching team gets:

- Main effect of direction: F(1, 194) = 0.66
- Main effect of station: F(3, 194) = 23.35
- Interaction between direction and station: F(3, 194) = 16.28

Two of those match the paper. One is close but not the same. Keep going; the bonus question at the end asks you to work out why.

Partial eta squared

Partial eta squared is the share of variance an effect accounts for, once the other effects are set aside:

$$\eta_p^2 = \frac{SS_{\text{effect}}}{SS_{\text{effect}} + SS_{\text{residual}}}$$

`tidy()` from broom turns the ANOVA table into a tibble, with one row per term and the sums of squares in the `sumsq` column. That means we can compute η_p^2 with ordinary tidyverse verbs.

```
anova_table <- tidy(mod)
anova_table
```

```
# A tibble: 4 x 6
  term                  df   sumsq meansq statistic   p.value
  <chr>              <dbl>   <dbl>  <dbl>     <dbl>     <dbl>
1 DIRECTION              1   0.713  0.713     0.664  4.16e- 1
2 STN_NAME               3  75.2   25.1      23.3    6.01e-13
3 DIRECTION:STN_NAME     3  52.4   17.5      16.3    1.77e- 9
4 Residuals            194 208.     1.07     NA     NA
```

```r
ss_residual <- anova_table |>
  filter(term == "Residuals") |>
  pull(sumsq)

anova_table |>
  filter(term != "Residuals") |>
  mutate(partial_eta_sq = sumsq / (sumsq + ss_residual)) |>
  select(term, df, statistic, p.value, partial_eta_sq)
```

```
# A tibble: 3 x 5
  term                 df statistic  p.value partial_eta_sq
  <chr>             <dbl>     <dbl>    <dbl>          <dbl>
1 DIRECTION             1     0.664 4.16e- 1        0.00341
2 STN_NAME              3    23.3   6.01e-13        0.265
3 DIRECTION:STN_NAME    3    16.3   1.77e- 9        0.201
```

Do your p^2 values round to the paper's .27 (station) and .20 (interaction)?

Station-by-station comparisons

Next the authors compare eastbound and westbound travelers at each station separately. A one-way ANOVA with a single two-level predictor is the same test as a t-test ($F = t^2$). With only one effect in the model, p^2 is just SS_direction / SS_total.

St. George

```r
# 1. Keep only the St. George ratings
st_george <- d |>
  filter(STN_NAME == "STG")

# 2. Which group rated St. George as farther away, eastbound or westbound?
#    No statistics yet, just the mean rating in each direction.
st_george |>
  group_by(DIRECTION) |>
  summarize(mean_distance = mean(DISTANCE), n = n())
```

```
# A tibble: 2 x 3
  DIRECTION mean_distance     n
  <fct>             <dbl> <int>
1 EAST               2.77    26
2 WEST               1.64    25
```

```r
# 3. One-way ANOVA: distance predicted by direction
st_george_mod <- aov(DISTANCE ~ DIRECTION, data = st_george)
tidy(st_george_mod)
```

```
# A tibble: 2 x 6
  term         df sumsq meansq statistic    p.value
  <chr>     <dbl> <dbl>  <dbl>     <dbl>      <dbl>
1 DIRECTION     1  16.3 16.3        18.8  0.0000723
2 Residuals    49  42.4  0.865      NA   NA
```

```r
# 4. Partial eta squared (reuse your code from above)
st_george_table <- tidy(st_george_mod)
st_george_table
```

```
# A tibble: 2 x 6
  term         df sumsq meansq statistic    p.value
  <chr>     <dbl> <dbl>  <dbl>     <dbl>      <dbl>
1 DIRECTION     1  16.3 16.3        18.8  0.0000723
2 Residuals    49  42.4  0.865      NA   NA
```

```r
st_george_resid <- st_george_table |>
  filter(term == "Residuals") |>
  pull(sumsq)

st_george_table |>
  filter(term != "Residuals") |>
  mutate(partial_eta_sq = sumsq / (sumsq + st_george_resid)) |>
  select(term, df, statistic, p.value, partial_eta_sq)
```

```
# A tibble: 1 x 5
  term         df statistic   p.value partial_eta_sq
  <chr>     <dbl>     <dbl>     <dbl>          <dbl>
1 DIRECTION     1      18.8 0.0000723          0.277
```

Spadina, Bloor-Yonge, and Sherbourne

Repeat the four steps for the other three stations. Copying the St. George chunk three times works. If you want to try something tidier, the **Going further** section below does all four stations in one pipeline.

```r
# 1. Keep only the Spadina ratings
spadina <- d |>
  filter(STN_NAME == "SPAD")

# 2. Which group rated Spadina as farther away, eastbound or westbound?
#    No statistics yet, just the mean rating in each direction.
spadina |>
  group_by(DIRECTION) |>
  summarize(mean_distance = mean(DISTANCE), n = n())
```

```
# A tibble: 2 x 3
  DIRECTION mean_distance     n
  <fct>             <dbl> <int>
1 EAST               3.65    26
2 WEST               2.64    25
```

```r
# 3. One-way ANOVA: distance predicted by direction
spadina_mod <- aov(DISTANCE ~ DIRECTION, data = spadina)
tidy(spadina_mod)
```

```
# A tibble: 2 x 6
  term         df sumsq meansq statistic  p.value
  <chr>     <dbl> <dbl>  <dbl>     <dbl>    <dbl>
1 DIRECTION     1  13.1  13.1       12.0  0.00113
2 Residuals    49  53.6   1.09      NA   NA
```

```r
# 4. Partial eta squared (reuse your code from above)
spadina_table <- tidy(spadina_mod)
spadina_table
```

```
# A tibble: 2 x 6
  term         df sumsq meansq statistic  p.value
  <chr>     <dbl> <dbl>  <dbl>     <dbl>    <dbl>
1 DIRECTION     1  13.1  13.1       12.0  0.00113
2 Residuals    49  53.6   1.09      NA   NA
```

```r
spadina_resid <- spadina_table |>
  filter(term == "Residuals") |>
  pull(sumsq)
```

```r
spadina_table |>
  filter(term != "Residuals") |>
  mutate(partial_eta_sq = sumsq / (sumsq + spadina_resid)) |>
  select(term, df, statistic, p.value, partial_eta_sq)
```

```
# A tibble: 1 x 5
  term         df statistic p.value partial_eta_sq
  <chr>     <dbl>     <dbl>   <dbl>          <dbl>
1 DIRECTION     1      12.0 0.00113          0.196
```

```r
# 1. Keep only the Bloor-Yonge ratings
bloor_yonge <- d |>
  filter(STN_NAME == "B-Y")

# 2. Which group rated B-Y as farther away, eastbound or westbound?
#    No statistics yet, just the mean rating in each direction.
bloor_yonge |>
  group_by(DIRECTION) |>
  summarize(mean_distance = mean(DISTANCE), n = n())
```

```
# A tibble: 2 x 3
  DIRECTION mean_distance     n
  <fct>             <dbl> <int>
1 EAST               1.61    23
2 WEST               2.19    26
```

```r
# 3. One-way ANOVA: distance predicted by direction
bloor_yonge_mod <- aov(DISTANCE ~ DIRECTION, data = bloor_yonge)
tidy(bloor_yonge_mod)
```

```
# A tibble: 2 x 6
  term         df sumsq meansq statistic p.value
  <chr>     <dbl> <dbl>  <dbl>     <dbl>   <dbl>
1 DIRECTION     1  4.16   4.16      3.95  0.0528
2 Residuals    47 49.5    1.05     NA    NA
```

```r
# 4. Partial eta squared (reuse your code from above)
bloor_yonge_table <- tidy(bloor_yonge_mod)
bloor_yonge_table
```

```
# A tibble: 2 x 6
  term         df sumsq meansq statistic p.value
  <chr>     <dbl> <dbl>  <dbl>     <dbl>   <dbl>
1 DIRECTION     1  4.16   4.16      3.95  0.0528
2 Residuals    47 49.5    1.05     NA    NA
```

```
bloor_yonge_resid <- bloor_yonge_table |>
  filter(term == "Residuals") |>
  pull(sumsq)

bloor_yonge_table |>
  filter(term != "Residuals") |>
  mutate(partial_eta_sq = sumsq / (sumsq + st_george_resid)) |>
  select(term, df, statistic, p.value, partial_eta_sq)
```

```
# A tibble: 1 x 5
  term         df statistic p.value partial_eta_sq
  <chr>     <dbl>     <dbl>   <dbl>          <dbl>
1 DIRECTION     1      3.95  0.0528         0.0893
```

```
# 1. Keep only the Sherbourne ratings
sherbourne <- d |>
  filter(STN_NAME == "SHER")

# 2. Which group rated Spadina as farther away, eastbound or westbound?
#    No statistics yet, just the mean rating in each direction.
sherbourne |>
  group_by(DIRECTION) |>
  summarize(mean_distance = mean(DISTANCE), n = n())
```

```
# A tibble: 2 x 3
  DIRECTION mean_distance     n
  <fct>             <dbl> <int>
1 EAST               2.77    26
2 WEST               4       25
```

```
# 3. One-way ANOVA: distance predicted by direction
sherbourne_mod <- aov(DISTANCE ~ DIRECTION, data = sherbourne)
tidy(sherbourne_mod)
```

```
# A tibble: 2 x 6
  term         df sumsq meansq statistic   p.value
  <chr>     <dbl> <dbl>  <dbl>     <dbl>     <dbl>
1 DIRECTION     1  19.3  19.3       15.1  0.000305
2 Residuals    49  62.6   1.28      NA   NA
```

```
# 4. Partial eta squared (reuse your code from above)
sherbourne_table <- tidy(sherbourne_mod)
sherbourne_table
```

```
# A tibble: 2 x 6
  term         df sumsq meansq statistic   p.value
  <chr>     <dbl> <dbl>  <dbl>     <dbl>     <dbl>
1 DIRECTION     1  19.3  19.3       15.1  0.000305
2 Residuals    49  62.6   1.28      NA   NA
```

```
sherbourne_resid <- sherbourne_table |>
  filter(term == "Residuals") |>
  pull(sumsq)

sherbourne_table |>
  filter(term != "Residuals") |>
  mutate(partial_eta_sq = sumsq / (sumsq + sherbourne_resid)) |>
  select(term, df, statistic, p.value, partial_eta_sq)
```

```
# A tibble: 1 x 5
  term         df statistic  p.value partial_eta_sq
  <chr>     <dbl>     <dbl>    <dbl>          <dbl>
1 DIRECTION     1      15.1 0.000305          0.236
```

Fill in this table as you go:

Station	Paper p	Your p	Paper p^2	Your p^2
St. George	$< .001$	$< .001$	.28	.28
Spadina	.001	.001	.20	.20
Bloor-Yonge	.053	.053	.08	.09
Sherbourne	$< .001$	$< .001$	.24	.24

Step 5: Reflection

Were you able to reproduce the results you attempted to reproduce? If not, which parts could you not reproduce?

Yes, except for reproducing the main effect and the p value for Bloor-Yonge for some reason.

How difficult was it to reproduce the results?

Relatively difficult

What made it hard, and what made it easy? Think about the data files, the variable names, and how much the paper told you about the analysis.

I'm not familiar with R, so it was difficult due to unfamiliarity the syntax. I also think the methods + results section was pretty short, so it felt more like I was stabbing at things over and over again to try to figure it out.

Bonus questions

1. Why is the station F different? Your F for station (23.35) is close to the paper's (24.10) but not equal, while the interaction matches exactly. Try refitting the model with the predictors in the opposite order, `DISTANCE ~ STN_NAME * DIRECTION`. Does anything change? Look back at your cell counts. What does that suggest about how the authors' software (SPSS) and R's `aov()` each divide up the variance? `docs/anova-in-r.md` explains it, but try first.

2. Orientation or direction? The paper describes the factor as "orientation: toward, away from". The dataset has `DIRECTION`: east or west. Are those the same variable? If not, which one did the authors actually put in their ANOVA? Make an `orientation` column with `if_else()` and refit the model with it to check.

3. Reproduce Figure 1. Plot the mean distance rating for each station, split by direction, with error bars of ± 1 standard error, as in the paper. `stat_summary(fun.data = mean_se)` will draw both the means and the error bars for you.

Going further

Splitting, modeling, and tidying once per station can all happen in one pipeline. `nest()` packs each station's rows into its own small data frame, and `map()` runs a function on each one:

```
d |>
  nest(.by = STN_NAME) |>
  mutate(model = map(data, \(df) aov(DISTANCE ~ DIRECTION, data = df)),
         results = map(model, tidy)) |>
  unnest(results)
```

Once that runs, add the p^2 column.